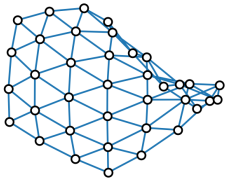
Graph Drawing and Tutte's Theorem

Ian Chen



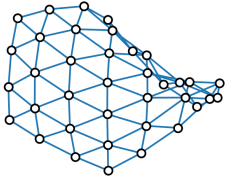
Graphs



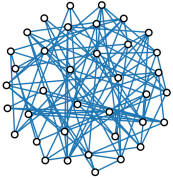
Spring



Graphs



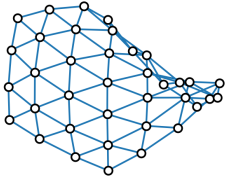
Spring



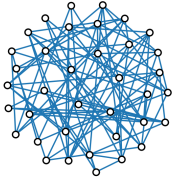
ARF



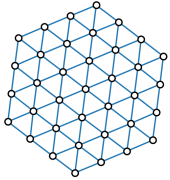
Graphs



Spring



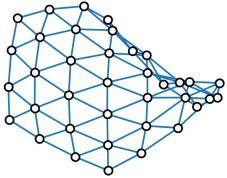
ARF



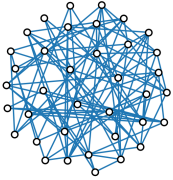
Kamada–Kawai



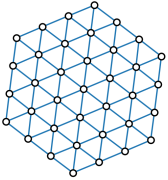
Graphs



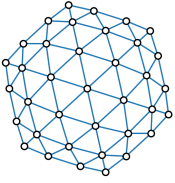
Spring



ARF



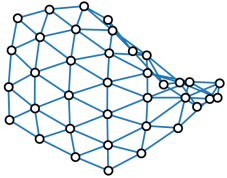
Kamada–Kawai



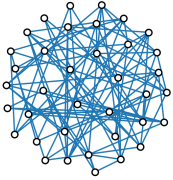
Spectral



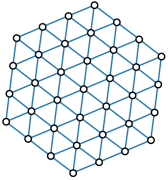
Graphs



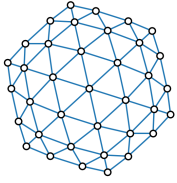
Spring



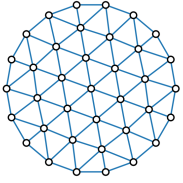
ARF



Kamada–Kawai



Spectral



Tutte



Outline

Overview
- Tutte
- Other force-directed

Tutte's Theorem



Overview

- Tutte
- Other force-directed

Tutte's Theorem



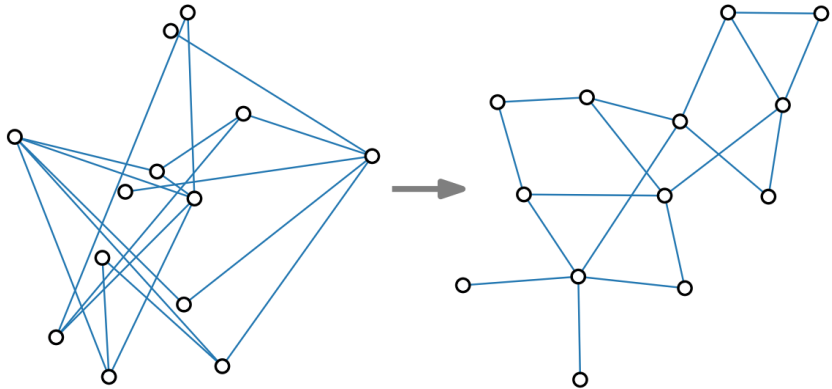
Tutte's Spring Embedding

- Graph $G = (V, E)$, embedding $p : V \to \mathcal{R}^2$ (denote $\boldsymbol{p}_v = p(v)$).



Tutte's Spring Embedding

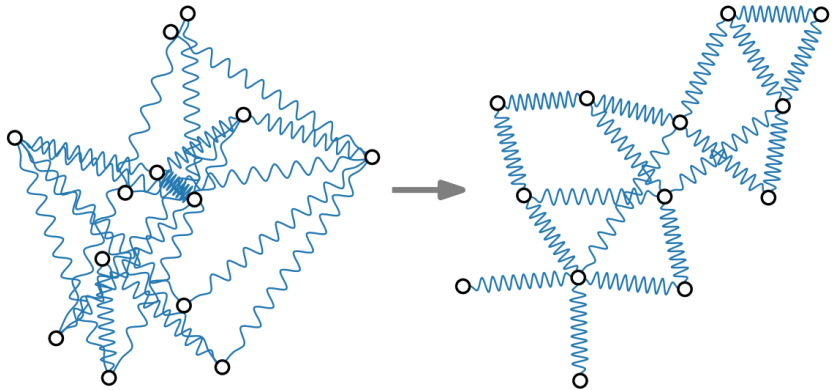
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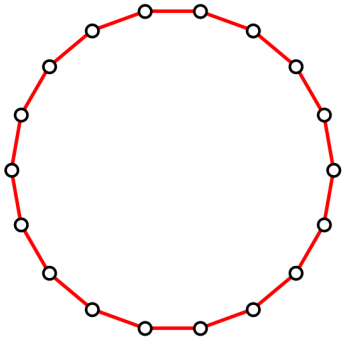
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Tutte's Spring Embedding (cont.)

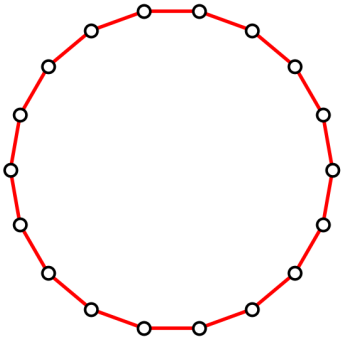
- boundary S positioned at convex polygon; free vertices $V \setminus S$

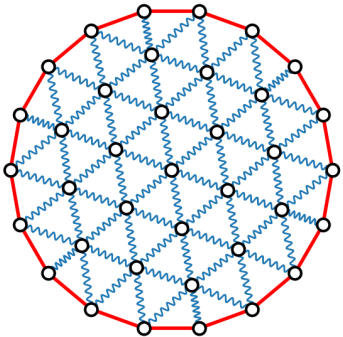




Tutte's Spring Embedding (cont.)

- boundary S positioned at convex polygon; free vertices $V \setminus S$







Tutte's Spring Embedding (cont.)

- Equilibrium condition

$$\sum_{v \in N(u)} k_{uv}(\boldsymbol{p}_v - \boldsymbol{p}_u) = \mathbf{0} \qquad \forall u \notin S$$



Tutte's Spring Embedding (cont.)

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$$\min \sum_{u \notin S} \sum_{v \in N(u)} \frac{1}{2} k_{uv} \|\boldsymbol{p}_v - \boldsymbol{p}_u\|_2^2$$



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- Newton-Raphson / Gaussian-Elimination

$$\boldsymbol{F}_u^t = -\deg(u)\boldsymbol{p}_u^t + \sum_{v \in N(u)} k_{uv}\boldsymbol{p}_v^t$$



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$$\boldsymbol{p}_u^{t+1} = \boldsymbol{p}_u^t + \frac{1}{\deg(u)}\boldsymbol{F}_u^t$$



Tutte's Spring Embedding (cont.)

- Theorem: (Whitney '33) every 3-connected planar graph has a unique embedding[1]



[1] up to homeomorphism, on the sphere

Tutte's Spring Embedding (cont.)

- Theorem: (Whitney '33) every 3-connected planar graph has a unique embedding[1]
- Theorem: (Tutte '63) Tutte embedding is strongly-convex embedding (if G 3-connected).



[1] up to homeomorphism, on the sphere

Kamada-Kawai layout

- spring equilibrium based on graph distances d_{uv}



Kamada-Kawai layout

- spring equilibrium based on graph distances d_{uv}
- spring stiffness k, scale L

$$f_{uv} = -\frac{k}{d_{uv}^2}(\|\boldsymbol{p}_v - \boldsymbol{p}_u\| - Ld_{uv})$$



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Anti-gravity drawing

- scale L, anti-gravity around vertices (repulsion)



Anti-gravity drawing

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Overview
Tutte
Other force-directed

Tutte's Theorem



Tutte's Spring Embedding (cont.)

- Theorem: (Whitney '33) every 3-connected planar graph has a unique embedding[2]



[2] up to homeomorphism, on the sphere

Tutte's Spring Embedding (cont.)

- Theorem: (Whitney '33) every 3-connected planar graph has a unique embedding[2]
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[2] up to homeomorphism, on the sphere

Proof outline

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  - $\Longrightarrow$ strictly-convex faces
  - $\Longrightarrow$ no degenerate vertices (embedding injective)
- Points are in exactly one face
  - $\Longrightarrow$ planar embedding



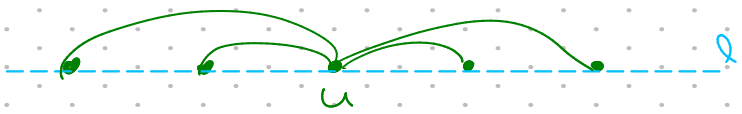
No degenerate Neighborhoods

Equilibrium condition:

$$\sum_v (p_v - p_u) = 0$$



No degenerate Neighborhoods

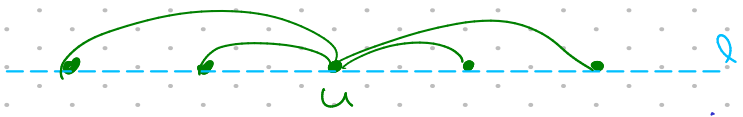


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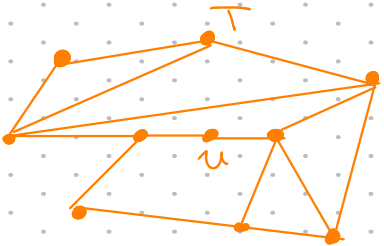
Σ

No degenerate Neighborhoods



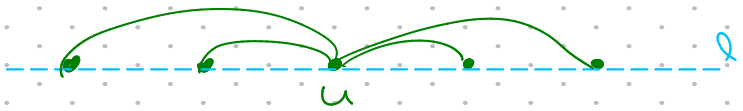
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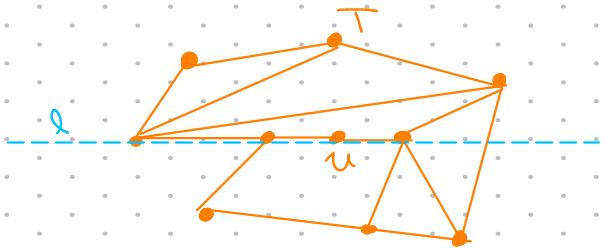
Tutte Embedding of G.

No degenerate Neighborhoods



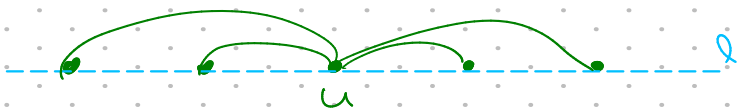
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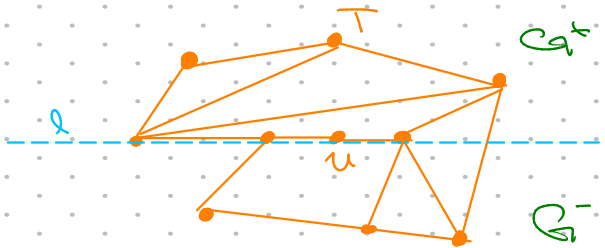
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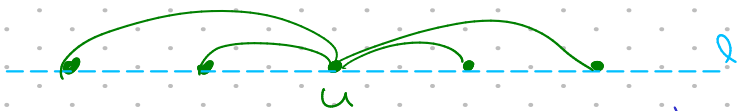
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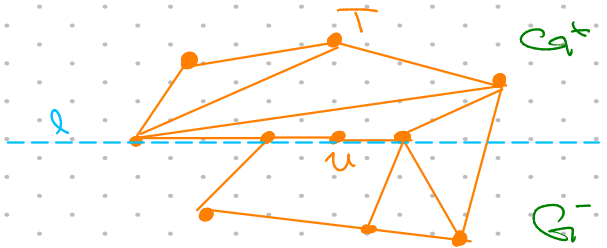
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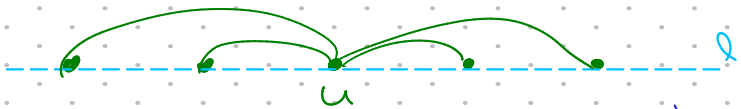
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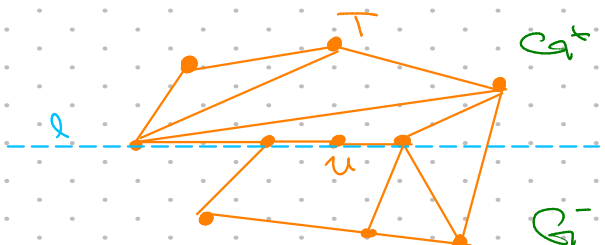
obs: every vertex has nondegenerating path to τ (by height)



Σ Tutte Embedding of G.

No degenerate Neighborhoods





Tutte Embedding of G.

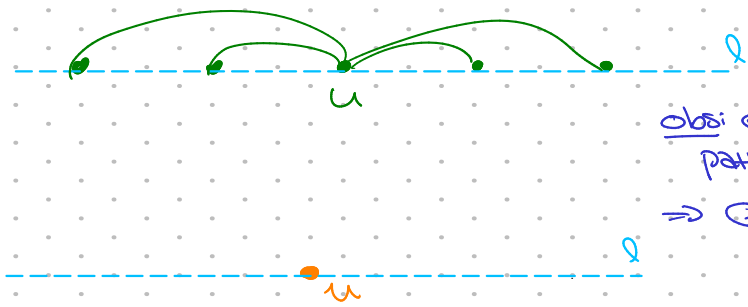
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$\Rightarrow G^+, G^-$ connected

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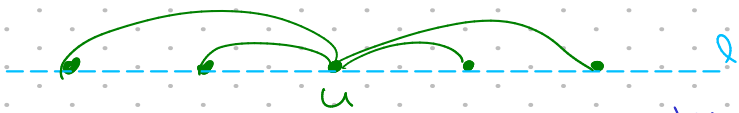
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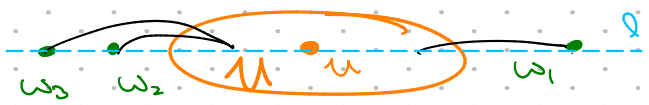
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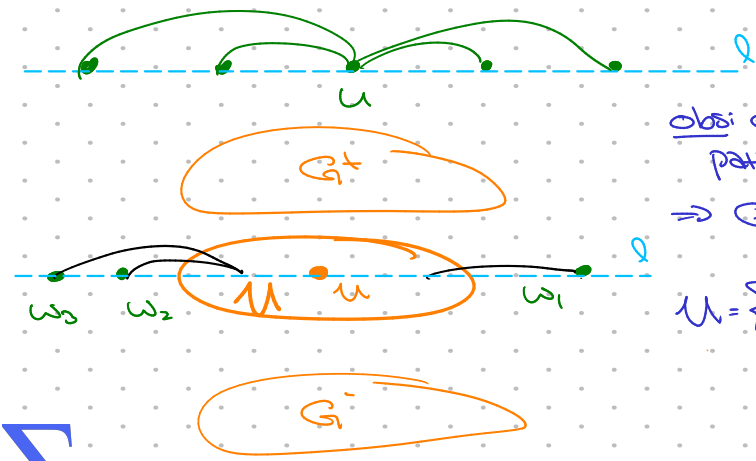
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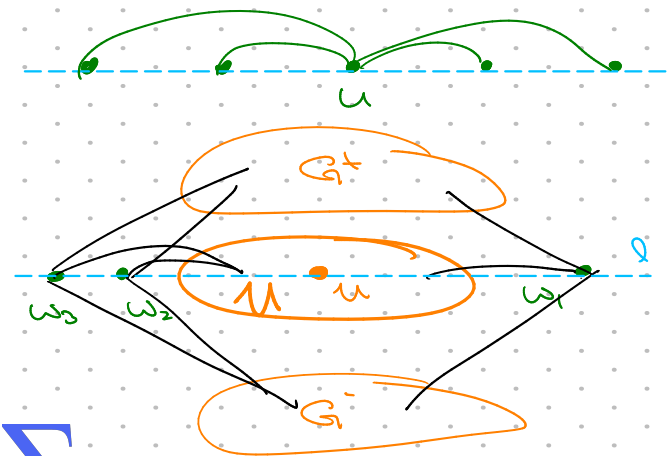
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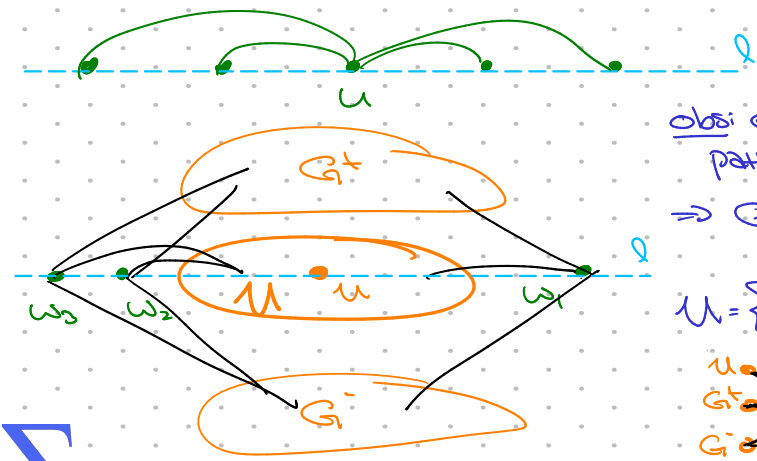
obs: every vertex has nondecreasing path to T (by height)

$\Rightarrow G^+, G^-$ connected

$U = \{ v \in V : N(v) \text{ on } \ell \}$

$\sum$ assuming $N(u)$ is degenerate

No degenerate Neighborhoods



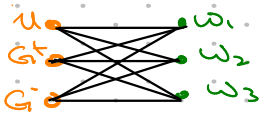
Equilibrium condition:

$$\sum_v (p_v - p_u) = 0$$

Obs: every vertex has nondecreasing path to T (by height)

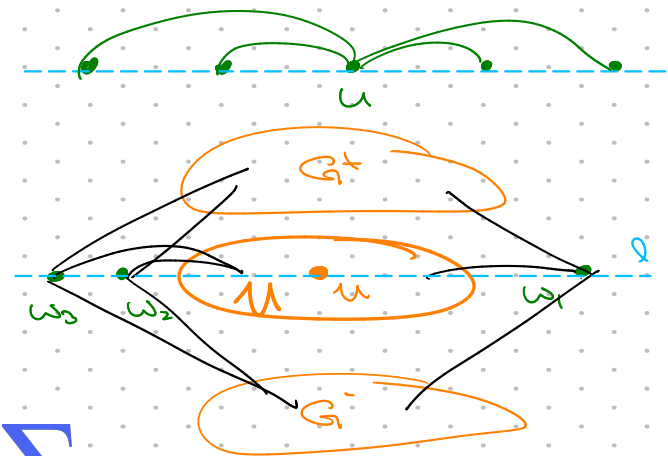
$\Rightarrow G^+, G^-$ connected

$U = \{ v \in V : N(v) \text{ on } \ell \}$



Σ assuming $N(u)$ is degenerate

No degenerate Neighborhoods



Σ assuming N(u) is degenerate

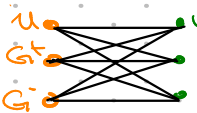
Equilibrium condition:

$$\sum_v (p_v - p_u) = 0$$

Obs: every vertex has nondecreasing path to T (by height)

⇒ G^+, G^- connected

$U = \{ v \in V : N(v) \text{ on } \ell \}$



$K_{3,3}$ minor ⇒ not planar

Proof outline

- No degenerate neighborhoods (not all neighborhood lies on a line)



Proof outline

- No degenerate neighborhoods (not all neighborhood lies on a line)
  - $\Longrightarrow$ every free vertex has neighbors strictly on both sides of any line



Proof outline

- No degenerate neighborhoods (not all neighborhood lies on a line)
  - $\Longrightarrow$ every free vertex has neighbors strictly on both sides of any line
  - $\Longrightarrow$ all free vertices inside the outer polygon



Split-Focus Lemma

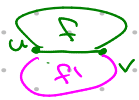


Equilibrium Condition:

$$\sum_v (p_v - p_u) = 0$$

Σ

Split-Faces Lemma



Equilibrium Condition:

$$\sum_v (p_v - p_u) = 0$$

Σ

Split-Faces Lemma:



f and f' are on (strictly) separate sides of l

Equilibrium Condition:

$\sum_v (p_v - p_u) = 0$



Split-Faces Lemma:



f and f' are on (strictly) separate sides of l

Equilibrium Condition:

$\sum_v (p_v - p_u) = 0$



$\sum$ position vertices to any plane always

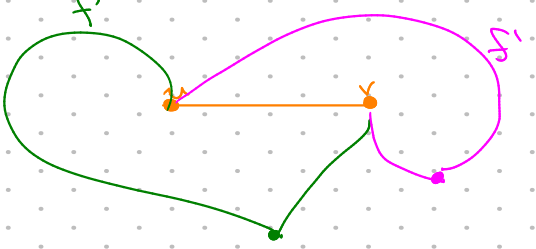
Split-Faces Lemma:



f and f' are on (strictly) separate sides of ℓ

Equilibrium Condition:

$\sum_v (p_v - p_u) = 0$



$\sum$ position vertices to any plane cancel

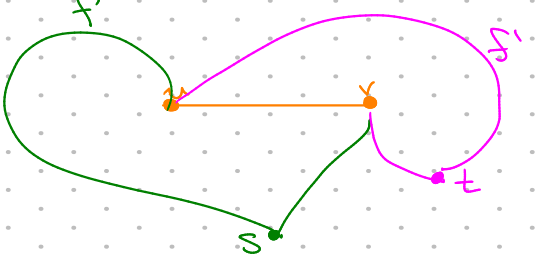
Split-Faces Lemma:



f and f' are on (strictly) separate sides of ℓ

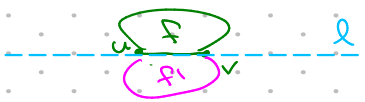
Equilibrium Condition:

$\sum_v (p_v - p_u) = 0$



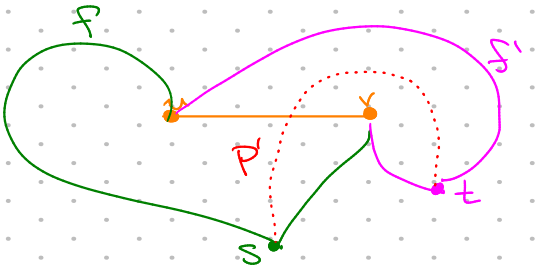
$\sum$ position vertices to any plane displacing

Split-Faces Lemma:



f and f' are on (strictly) separate sides of l

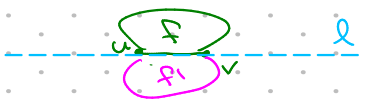
Equilibrium Condition:
$\sum_v (p_v - p_u) = 0$



① P' curve in $\mathbb{R}^2$ from s-t contained in f ∪ f'

Σ position vertices to any point cancels

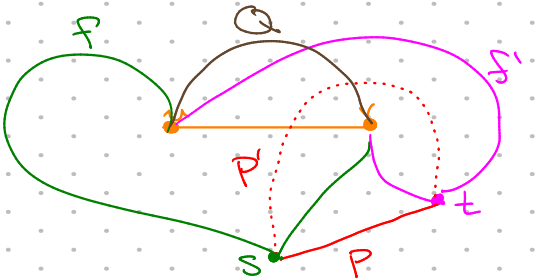
Split-Faces Lemma



f and f' are on (strictly) separate sides of l

Equilibrium Condition:

$\sum_v (p_v - p_u) = 0$



① P' curve in $\mathbb{R}^2$ from s-t contained in f, f'

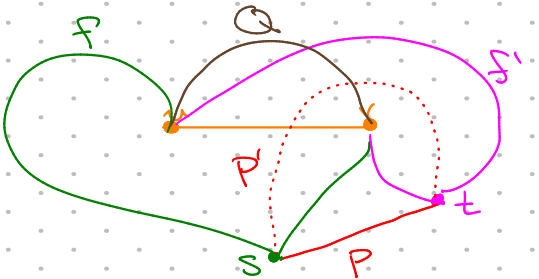
Σ position vertices to any place allowing

Split-Faces Lemma:



f and f' are on (strictly) separate sides of l

Equilibrium Condition:
$\sum_v (p_v - p_u) = 0$

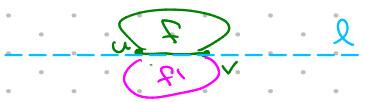


(1) P' curve in $\mathbb{R}^2$ from s-t contained in f, f'

(2) Any Q (u→v) must intersect P (s→t)

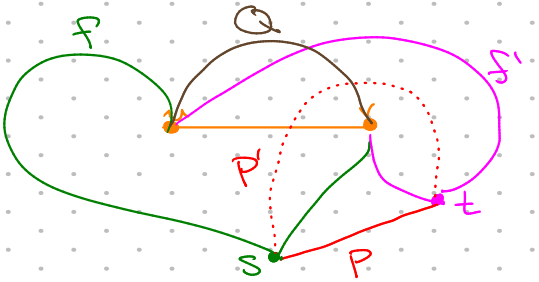
$\sum$ position vertices to any point displacing

Split-Faces Lemma:



f and f' are on (strictly) separate sides of l

Equilibrium Condition:
$\sum_v (p_v - p_u) = 0$



① P' curve in $\mathbb{R}^2$ from s-t contained in f, f'

② Any Q (u→v) must intersect P (s→t)

③ If s, t on same side:
⇒ choose P in V^-
⇒ choose Q in V^+

$\sum$ position vertices to any point cancels

Proof outline

- Split-faces lemma (adjacent faces strictly separated by a line)



Proof outline

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  - $\Longrightarrow$ strictly-convex faces



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Proof outline

- Split-faces lemma (adjacent faces strictly separated by a line)
  - $\Longrightarrow$ strictly-convex faces
  - $\Longrightarrow$ no degenerate vertices (embedding injective)
- Points are in exactly one face



Proof outline

- Split-faces lemma (adjacent faces strictly separated by a line)
  - $\Longrightarrow$ strictly-convex faces
  - $\Longrightarrow$ no degenerate vertices (embedding injective)
- Points are in exactly one face
  - $\Longrightarrow$ planar embedding

